

Evolution of Fluid Flow Analysis

The study of fluid dynamics has evolved through both experimental and theoretical approaches, focusing on understanding how fluids behave in different environments and developing fundamental principles and relationships.

Objectives

The goal is to show mathematical equations that describe how reservoir fluids move under various conditions, considering different fluid categories, types of flow conditions, the shape and structure of reservoirs, and the quantity of moving fluids.

TYPES OF FLUIDS

The isothermal compressibility coefficient is essentially the controlling factor in identifying the type of the reservoir fluid. In general, reservoir fluids are classified into three groups:

- Incompressible fluids
- Slightly compressible fluids
- Compressible fluids

the isothermal compressibility coefficient c is described mathematically by the following two equivalent expressions:

In terms of fluid volume:

$$c = \frac{-1}{V} \frac{\partial V}{\partial p} \quad (1)$$

In terms of fluid density:

$$c = \frac{-1}{V} \frac{\partial \rho}{\partial p} \quad (2)$$

where V and ρ are the volume and density of the fluid, respectively.

Incompressible fluids

These are fluids whose volume or density does not change with pressure. Perfectly incompressible fluids are theoretical and do not exist.

$$\frac{\partial V}{\partial p} = 0$$
$$\frac{\partial \rho}{\partial p} = 0$$

Incompressible fluids do not exist; this behavior, however, may be assumed in some cases to simplify the derivation and the final form of many flow equations.

Slightly compressible fluids

Slightly Compressible Fluids

These fluids exhibit minor variations in volume or density with changes in pressure. Their volume and density can be described by the equations:

$$-c \int_{P_{ref}}^P dP = \int_{V_{ref}}^V \frac{dV}{V}$$

$$e^{c(P_{ref} - P)} = \frac{V}{V_{ref}}$$

$$V = V_{ref} e^{c(P_{ref} - P)} \dots\dots\dots(3)$$

where p = pressure, psia

V = volume at pressure p, ft³

p_{ref} = initial (reference) pressure, psia

V_{ref} = fluid volume at initial (reference) pressure, psia

The e^x may be represented by a series expansion as:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} \dots\dots\dots(4)$$

Because the exponent x [which represents the term c (p_{ref}–p)] is very small, the e^x term can be approximated by truncating Equation 4 to:

$$e^x = 1 + x \dots\dots\dots(5)$$

Combining Equation 5 with Equation 3 gives:

$$V = V_{ref} [1 + c (p_{ref} - p)] \dots\dots\dots(6)$$

A similar derivation is applied to Equation 2 to give

$$\rho = \rho_{ref} [1 - c (p_{ref} - p)] \dots\dots\dots(7)$$

where V = volume at pressure p

ρ = density at pressure p

V_{ref} = volume at initial (reference) pressure p_{ref}

ρ_{ref} = density at initial (reference) pressure p_{ref}

It should be pointed out that crude oil and water systems fit into this category

Compressible Fluids

These fluids undergo significant volume changes with pressure variations. Examples include all gases, and their behavior can be described using a full series expansion for calculations, as given by Equation 4

The isothermal compressibility of any compressible fluid is described by the following expression:

$$C_g = \frac{1}{p} - \frac{1}{z} \left(\frac{\partial z}{\partial p} \right)_T \dots\dots\dots(8)$$

Figures 1 and 2 show schematic illustrations of the volume and density changes as a function of pressure for the three types of fluids.

FLOW REGIMES

There are basically three types of flow regimes that must be recognized in order to describe the fluid flow behavior and reservoir pressure distribution as a function of time.

There are three flow regimes:

- Steady-state flow
- Unsteady-state flow
- Pseudosteady-state flow

Steady-State Flow

The flow regime is identified as a steady-state flow if the Pressure remains constant throughout the flow. Mathematically, this condition is expressed as:

$$\left(\frac{\partial p}{\partial t} \right)_i = 0 \dots\dots\dots(9)$$

The above equation states that the rate of change of pressure p with respect to time t at any location i is zero. In reservoirs, the steady-state flow condition can only occur when the reservoir is completely recharged and supported by strong aquifer or pressure maintenance operations.

Unsteady-State Flow

The unsteady-state flow (frequently called *transient flow*) is defined as the fluid flowing condition at which pressure changes over time and position. This definition suggests that the pressure derivative with respect to time is essentially a function of both position i and time t , thus

$$\left(\frac{\partial p}{\partial t} \right)_i = f(i, t) \dots \dots \dots (10)$$

Pseudo steady-State Flow

When the pressure at different locations in the reservoir is declining linearly as a function of time, i.e., at a constant declining rate, the flowing condition is characterized as the pseudosteady-state flow. Mathematically, this definition states that the rate of change of pressure with respect to time at every position is constant, or

$$\left(\frac{\partial p}{\partial t} \right)_i = \text{const} \tan t \dots \dots \dots (11)$$

It should be pointed out that the pseudosteady-state flow is commonly referred to as semisteady-state flow and quasisteady-state flow. Figure 1 shows a schematic comparison of the pressure declines as a function of time of the three flow regimes.

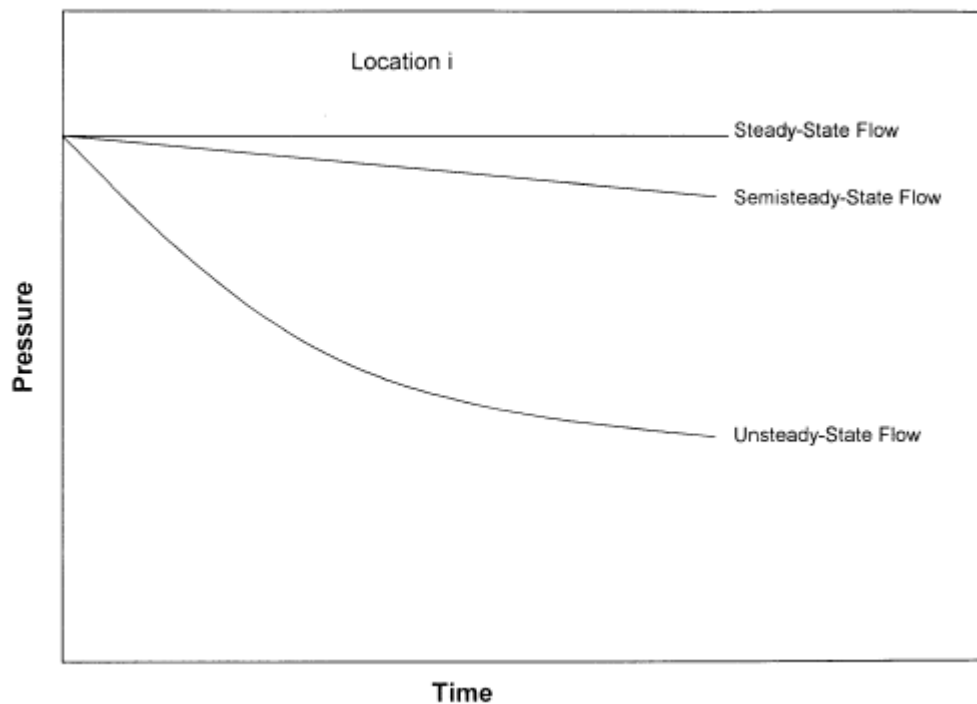


Figure 1 shows a schematic comparison of the pressure declines as a function of time of the three flow regimes.

RESERVOIR GEOMETRY

The shape of a reservoir has a significant effect on its flow behavior. Most reservoirs have irregular boundaries and a rigorous mathematical description of geometry is often possible only with the use of numerical simulators. For many engineering purposes, however, the actual flow geometry may be represented by one of the following flow geometries:

- Radial flow
- Linear flow
- Spherical and hemispherical flow

Radial Flow

In the absence of severe reservoir heterogeneities, flow into or away from a wellbore will follow radial flow lines from a substantial distance from the wellbore. Because fluids move toward the well from all directions and coverage at the wellbore, Figure 4 shows idealized flow lines and iso-potential lines for a radial flow system.

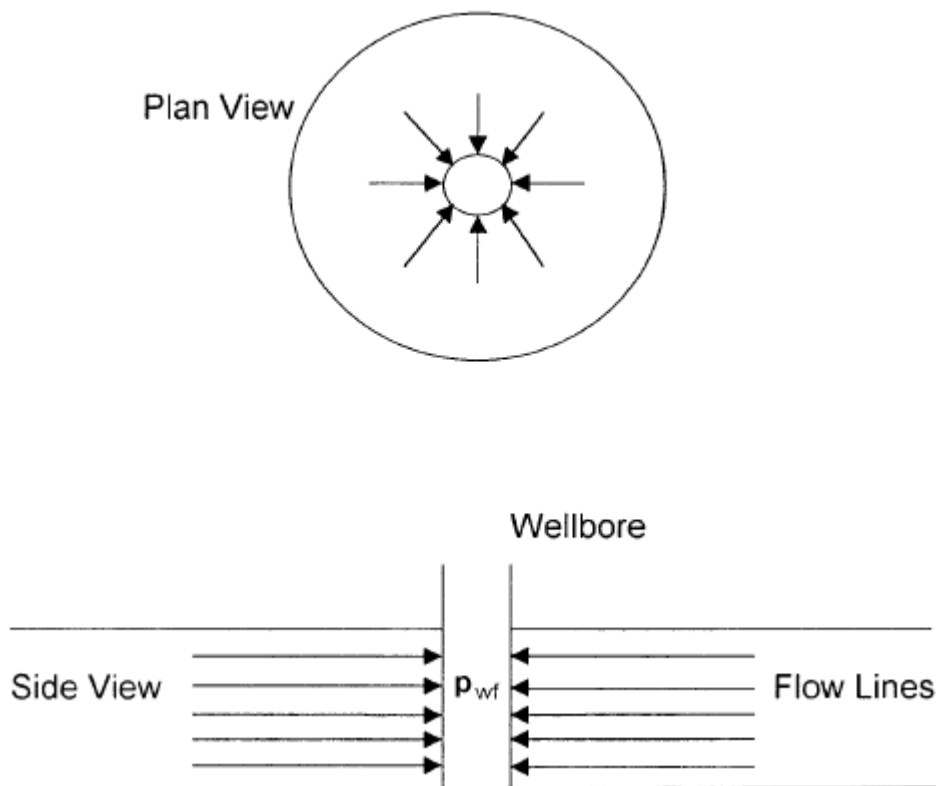


Figure 4. Radial flow into a wellbore.

Linear Flow

In the case of linear flow, Fluids flow in parallel paths in a single direction, such as into vertical hydraulic fractures. Figure 5 shows an idealized linear flow system. A common application of linear flow equations is the fluid flow into vertical hydraulic fractures as illustrated in Figure 6.

Spherical and Hemispherical Flow

Depending upon the type of wellbore completion configuration, it is possible to have a spherical or hemispherical flow near the wellbore. A well with a limited perforated interval could result in spherical flow in the vicinity of the perforations as illustrated in Figure 7. A well that only partially penetrates the pay zone, as shown in Figure 8, could result in hemispherical flow. The condition could arise where coning of bottom water is important.

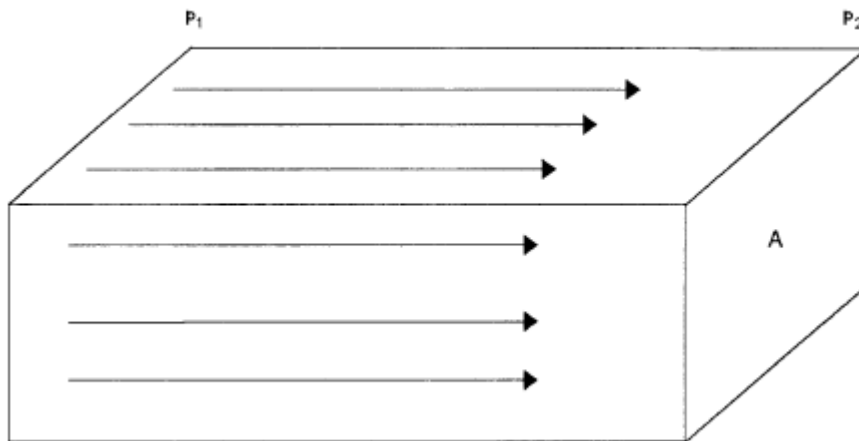


Figure 6. Linear flow.

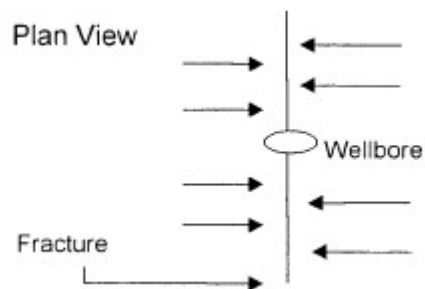


Figure 5. Ideal linear flow into vertical fracture.

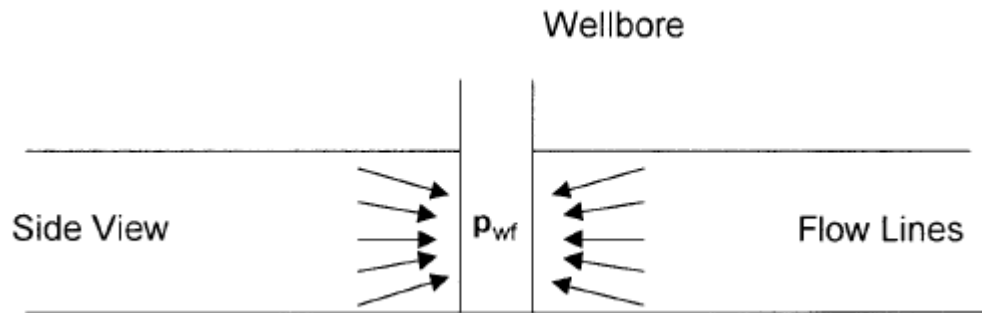


Figure 7. Spherical flow due to limited entry.

NUMBER OF FLOWING FLUIDS IN THE RESERVOIR

The mathematical expressions that are used to predict the volumetric performance and pressure behavior of the reservoir vary in forms and complexity depending upon the number of mobile fluids in the reservoir. There are generally three cases of flowing systems:

- Single-phase flow (oil, water, or gas)
- Two-phase flow (oil-water, oil-gas, or gas-water)
- Three-phase flow (oil, water, and gas)

The description of fluid flow and subsequent analysis of pressure data becomes more difficult as the number of mobile fluids increases.

FLUID FLOW EQUATIONS

The fluid flow equations that are used to describe the flow behavior in a reservoir can take many forms depending upon the combination of variables presented previously, (i.e., types of flow, types of fluids, etc.). By combining the conservation of mass equation with the transport equation (Darcy's equation) and various equations-of-state, the necessary flow equations can be developed. Since all flow equations to be considered depend on Darcy's Law, it is important to consider this transport relationship first.

Darcy's Law

The fundamental law of fluid motion in porous media is Darcy's Law. The mathematical expression developed by Henry Darcy in 1856 states the velocity of a homogeneous fluid in a porous medium is proportional to the pressure gradient and inversely proportional to the fluid viscosity. For a horizontal linear system, this relationship is:

$$v = \frac{q}{A} = - \frac{k}{\mu} \frac{dp}{dx} \dots\dots\dots(12)$$

v is the apparent velocity in centimeters per second and is equal to q/A , where q is the volumetric flow rate in cubic centimeters per second and A is total cross-sectional area of

the rock in square centimeters. In other words, A includes the area of the rock material as well as the area of the pore channels. The fluid viscosity, μ , is expressed in centipoise units, and the pressure gradient, dp/dx , is in atmospheres per centimeter, taken in the same direction as v and q . The proportionality constant, k , is the permeability of the rock expressed in Darcy units. The negative sign in Equation 6-12 is added because the pressure gradient is negative in the direction of flow as shown in Figure 6-9. For a horizontal-radial system, the pressure gradient is positive (see Figure 6-10) and Darcy's equation can be expressed in the following generalized radial form:

$$v = \frac{q_r}{A_r} = \frac{k}{\mu} \left(\frac{\partial p}{\partial r} \right)_r \dots\dots\dots(13)$$

where q_r = volumetric flow rate at radius r

A_r = cross-sectional area to flow at radius r

$(\partial p / \partial r)_r$ = pressure gradient at radius r

v = apparent velocity at radius r

The cross-sectional area at radius r is essentially the surface area of a cylinder. For a fully penetrated well with a net thickness of h , the crosssectional area A_r is given by:

$$A_r = 2 \pi r h$$

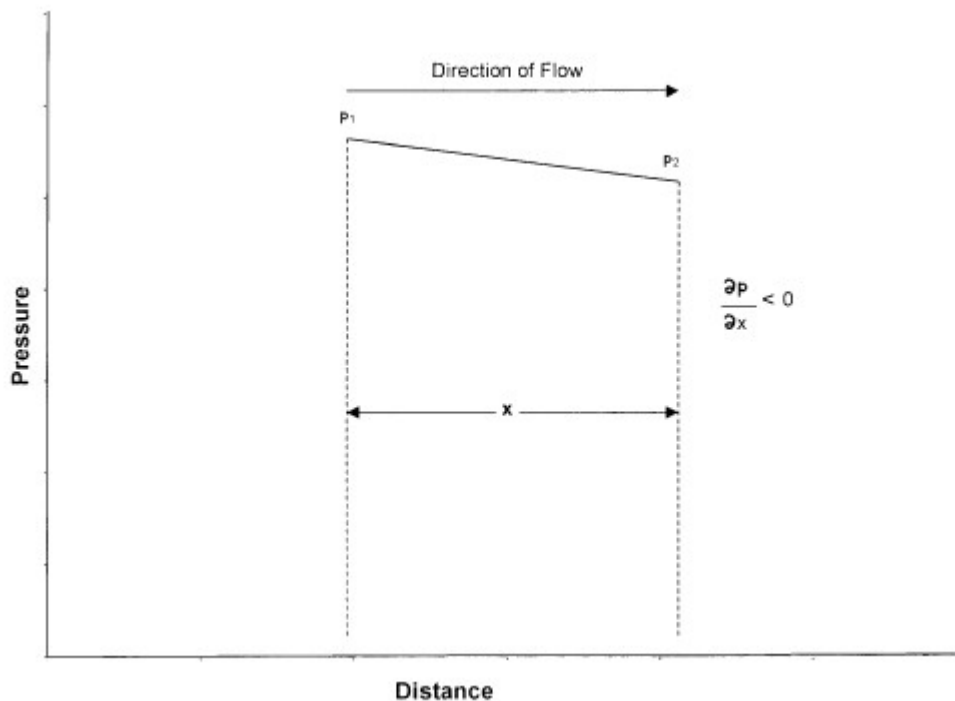


Figure 9. Pressure vs. distance in a linear flow.

Darcy's Law applies only when the following conditions exist:

- Laminar (viscous) flow
- Steady-state flow
- Incompressible fluids
- Homogeneous formation

For turbulent flow, which occurs at higher velocities, the pressure gradient increases at a greater rate than does the flow rate and a special modification of Darcy's equation is needed. When turbulent flow exists, the application of Darcy's equation can result in serious errors. Modifications for turbulent flow will be discussed later.

STEADY-STATE FLOW

As defined previously, steady-state flow represents the condition that exists when the pressure throughout the reservoir does not change with time. The applications of the steady-state flow to describe the flow behavior of several types of fluid in different reservoir geometries are presented to include:

- Linear flow of incompressible fluids
- Linear flow of slightly compressible fluids
- Linear flow of compressible fluids
- Radial flow of incompressible fluids
- Radial flow of slightly compressible fluids
- Radial flow of compressible fluids
- Multiphase flow

Linear Flow of Incompressible Fluids

In the linear system, it is assumed the flow occurs through a constant cross-sectional area A , where both ends are entirely open to flow. It is also assumed that no flow crosses the sides, top, or bottom as shown in Figure 11.

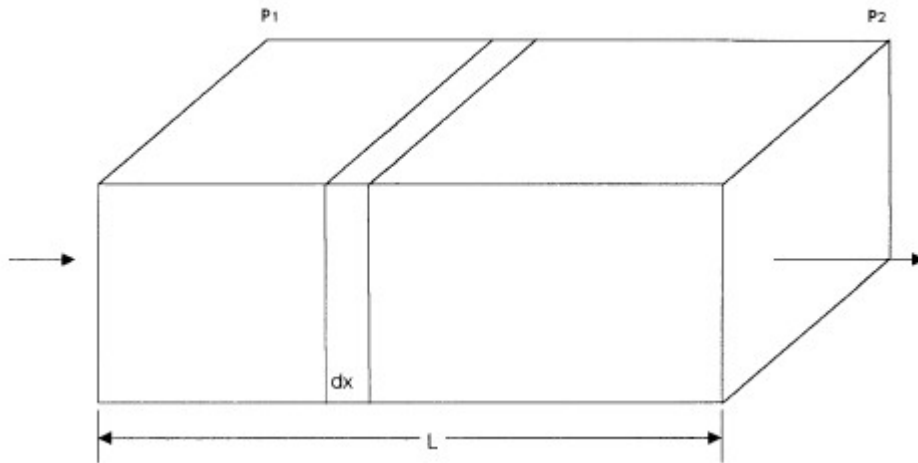


Figure 11. Linear flow model

If an incompressible fluid is flowing across the element dx , then the fluid velocity v and the flow rate q are constants at all points. The flow behavior in this system can be expressed by the differential form of

Darcy's equation, i.e., Equation 12. Separating the variables of Equation 12 and integrating over the length of the linear system gives:

$$\frac{q}{A} \int_0^L dx = - \frac{k}{\mu} \int_{p_1}^{p_2} dp$$

Or

$$q = \frac{kA(p_1 - p_2)}{\mu L}$$

It is desirable to express the above relationship in customary field units, or:

$$q = \frac{0.001127kA(p_1 - p_2)}{\mu L} \dots\dots\dots(14)$$

where q = flow rate, bbl/day
 k = absolute permeability, md
 p = pressure, psia
 μ = viscosity, cp
 L = distance, ft
 A = cross-sectional area, ft^2

Example 6-1

An incompressible fluid flows in a linear porous media with the following properties:

$L = 2000 \text{ ft}$; $h = 20'$; width = $300'$

$k = 100 \text{ md}$; $\phi = 15\%$; $\mu = 2 \text{ cp}$
 $p_1 = 2000 \text{ psi}$; $p_2 = 1990 \text{ psi}$

Calculate:

- Flow rate in bbl/day
- Apparent fluid velocity in ft/day
- Actual fluid velocity in ft/day

The difference in the pressure ($p_1 - p_2$) in Equation 14 is not the only driving force in a tilted reservoir. The gravitational force is the other important driving force that must be accounted for to determine the direction and rate of flow. The fluid gradient force (gravitational force) is always directed vertically downward while the force that results from an applied pressure drop may be in any direction. The force causing flow would be then the vector sum of these two. In practice, we obtain this result by introducing a new parameter, called fluid potential, which has the same dimensions as pressure, e.g., psi. Its symbol is Φ . The fluid potential at any point in the reservoir is defined as the pressure at that point less the pressure that would be exerted by a fluid head extending to an arbitrarily assigned datum level. Letting Δz_i be the vertical distance from a point i in the reservoir to this datum level.

$$\Phi_i = p_i - \left(\frac{\rho}{144} \right) \Delta z_i \dots\dots\dots(15)$$

where ρ is the density in lb/ft^3 . Expressing the fluid density in gm/cc in Equation 15 gives:

$$\Phi_i = p_i - 0.433 \gamma \Delta z_i \dots\dots\dots(16)$$

where Φ_i = fluid potential at point i , psi

p_i = pressure at point i , psi

Δz_i = vertical distance from point i to the selected datum level

ρ = fluid density, lb/ft^3

γ = fluid density, gm/cm^3

The datum is usually selected at the gas-oil contact, oil-water contact, or at the highest point in formation. In using Equations 15 or 16 to calculate the fluid potential Φ_i at location i , the vertical distance Δz_i is assigned as a positive value when the point i is below the datum level and as a negative when it is above the datum level, i.e.:

If point i is above the datum level:

$$\Phi = p_i + \left(\frac{\rho}{144} \right) \Delta z_i$$

And

$$\Phi = p_i - 0.433 \gamma \Delta z_i$$

If point i is below the datum level:

$$\Phi = p_i - \left(\frac{\rho}{144} \right) \Delta z_i$$

And

$$\Phi_i = p_i - 0.433 \gamma \Delta z_i$$

Applying the above-generalized concept to Darcy's equation (Equation 14) gives:

$$q = \frac{0.001127 k A (\Phi_1 - \Phi_2)}{\mu L} \dots \dots \dots (17)$$

It should be pointed out that the fluid potential drop ($\Phi_1 - \Phi_2$) is equal to the pressure drop ($p_1 - p_2$) only when the flow system is horizontal

Linear Flow of Slightly Compressible Fluids

Equation 6-6 describes the relationship that exists between pressure and volume for slightly compressible fluid, or:

$$V = V_{ref} [1 + c (p_{ref} - p)]$$

The above equation can be modified and written in terms of flow rate as:

$$q = q_{ref} [1 + c (p_{ref} - p)] \dots \dots \dots (18)$$

where q_{ref} is the flow rate at some reference pressure p_{ref} . Substituting the above relationship in Darcy's equation gives:

$$\frac{q}{A} = \frac{q_{ref} [1 + c (p_{ref} - p)]}{A} = -0.001127 \frac{k}{\mu} \frac{dp}{dx}$$

Separating the variables and arranging:

$$\frac{q_{ref}}{A} \int_0^L dx = -0.001127 \frac{k}{\mu} \int_{p_1}^{p_2} \left[\frac{dp}{1 + c (p_{ref} - p)} \right]$$

Integrating gives:

$$q_{ref} = \left[\frac{0.001127kA}{\mu cL} \right] \ln \left[\frac{1 + c(p_{ref} - p_2)}{1 + c(p_{ref} - p_1)} \right] \dots\dots\dots(19)$$

where q_{ref} = flow rate at a reference pressure p_{ref} , bbl/day

p_1 = upstream pressure, psi

p_2 = downstream pressure, psi

k = permeability, md

μ = viscosity, cp

c = average liquid compressibility, psi^{-1}

Selecting the upstream pressure p_1 as the reference pressure p_{ref} and substituting in Equation 6-19 gives the flow rate at Point 1 as:

$$q_1 = \left[\frac{0.001127kA}{\mu cL} \right] \ln [1 + c(p_1 - p_2)] \dots\dots\dots(20)$$

Choosing the downstream pressure p_2 as the reference pressure and substituting in Equation 19 gives:

$$q_2 = \left[\frac{0.001127kA}{\mu cL} \right] \ln \left[\frac{1}{1 + c(p_1 - p_2)} \right] \dots\dots\dots(21)$$

where q_1 and q_2 are the flow rates at point 1 and 2, respectively.

Linear Flow of Compressible Fluids (Gases)

For a viscous (laminar) gas flow in a homogeneous-linear system, the real-gas equation-of-state can be applied to calculate the number of gas moles n at pressure p , temperature T , and volume V :

$$n = \frac{pV}{zRT}$$

At standard conditions, the volume occupied by the above n moles is given by:

$$V_{sc} = \frac{nZ_{sc}RT_{sc}}{p_{sc}}$$

Combining the above two expressions and assuming $z_{sc} = 1$ gives:

$$\frac{pV}{zT} = \frac{P_{sc} V_{sc}}{T_{sc}}$$

Equivalently, the above relation can be expressed in terms of the flow rate as:

$$\frac{5.615pq}{zT} = \frac{P_{sc} Q_{sc}}{T_{sc}}$$

Rearranging

$$\left(\frac{P_{sc}}{T_{sc}} \right) \left(\frac{zT}{P} \right) \left(\frac{Q_{sc}}{5.615} \right) = q \dots \dots \dots (22)$$

where q = gas flow rate at pressure p in bbl/day

Q_{sc} = gas flow rate at standard conditions, scf/day

z = gas compressibility factor

T_{sc}, P_{sc} = standard temperature and pressure in °R and psia,
Respectively

Replacing the gas flow rate q with that of Darcy's Law, i.e., Equation 12, gives:

$$\frac{q}{A} = \left(\frac{P_{sc}}{T_{sc}} \right) \left(\frac{zT}{p} \right) \left(\frac{Q_{sc}}{5.615} \right) \left(\frac{1}{A} \right) = -0.001127 \frac{k}{\mu} \frac{dp}{dx}$$

The constant 0.001127 is to convert from Darcy's units to field units. Separating variables and arranging yields:

$$\left[\frac{q_{sc} P_{sc} T}{0.006328 K T_{sc} A} \right] \int_0^L dx = - \int_{p_1}^{p_2} \frac{p}{z \mu_g} dp$$

Assuming constant z and μ_g over the specified pressures, i.e., p_1 and p_2 , and integrating gives:

$$Q_{sc} = \frac{0.003164 T_{sc} A k (p_1^2 - p_2^2)}{P_{sc} T L z \mu_g}$$

where Q_{sc} = gas flow rate at standard conditions, scf/day

k = permeability, md

T = temperature, °R

μ_g = gas viscosity, cp

A = cross-sectional area, ft²

L = total length of the linear system, ft

Setting $p_{sc}=14.7$ psi and $T_{sc} = 520$ °R in the above expression gives:

$$Q_{sc} = \frac{0.111924Ak(p_1^2 - p_2^2)}{TLz\mu_g} \dots\dots\dots (23)$$

It is essential to notice that those gas properties z and μ_g are a very strong function of pressure, but they have been removed from the integral to simplify the final form of the gas flow equation. The above equation is valid for applications when the pressure < 2000 psi. The gas properties must be evaluated at the average pressure $\bar{p}$ – as defined below.

$$\bar{p} = \sqrt{\frac{p_1^2 + p_2^2}{2}} \dots\dots\dots (24)$$